FIRS Sampling Exam

Sampling Exercise

Planning for Zürichberg Post-Storm Remeasurement

The Zürichberg forest was inventoried the year before a major winter storm. The historical inventory contains measurements of stem number and basal area on field plots.

The forest owner now wants to know:

1. the total number of stems affected by the storm, and
2. the mean basal area affected by the storm.

Field remeasurement is expensive, so only a sample of the old plots can be remeasured. Your task is to propose **three different probability sampling designs**. For each design:

- explain how the sample will be selected and the reasoning for implementing it;
  - SRS without replacement
  - Two-stage sampling
  - cluster sampling
  - Stratification
  - etc...
- draw a realization of the sample to show to the owner;
- specify the estimator of total affected stems;
- specify the estimator of mean affected basal area;
- explain how the estimator will look and how uncertainty would be estimated;
- discuss advantages and disadvantages from the owner's perspective.

The owner doesn't know anything about sampling and will choose among your proposals so you will want to express yourself visually.

The dataset is loaded as follows:

```
library(forestinventory)
data(zberg)

zberg_forest <- zberg[!is.na(zberg$basal), ]
```

The package documentation describes `stem` as field-derived stem number and `basal` as field-derived basal area. It also provides `cluster`, `stade`, `melange`, and `couver` as potentially useful auxiliary information.

Estimation Exercise

This exercise is a continuation of the Monte Carlo integration example discussed in class. In that example, we considered estimating an integral using randomly generated points. Here, we consider a three-dimensional "local density surface" that we want to integrate over a two-dimensional domain.

The function is defined as follows:

```
Y <- function(x1, x2, true_beta = c(30, 13, -6, -4, 3, 2)) {
  Z_x <- cbind(
    1,
    x1,
    x2,
    x1^2,
    x1 * x2,
    x2^2
  )

  noise <- 6 * cos(pi * x1) * sin(2 * pi * x2)

  as.numeric(Z_x %*% true_beta + noise)
}

# Example
Y(.5, .5)
```

The domain of integration is

$$0 \leq x_1 \leq 3, \qquad 0 \leq x_2 \leq 4.$$

The surface can be visualized using:

```
library(plotly)

# Grid of x1 and x2 values
x1 <- seq(0, 3, length.out = 100)
x2 <- seq(0, 4, length.out = 100)

# Evaluate Y on the grid
z <- outer(x1, x2, Vectorize(Y))

# 3D surface
plot_ly(
  x = x1,
  y = x2,
  z = z
) |>
  add_surface() |>
  layout(
    scene = list(
      xaxis = list(title = "x1"),
      yaxis = list(title = "x2"),
      zaxis = list(title = "Y(x1, x2)")
    )
  )
```

We can obtain a numerical value for the integral using the `cubature` package:

```
library(cubature)
```

```
result <- adaptIntegrate(
  f = function(x) {Y(x[1], x[2])},
  lowerLimit = c(0, 0),
  upperLimit = c(3, 4)
)
result$integral
# 542
```

This value will be treated as the true value of the integral for purposes of this exercise.

We define the auxiliary variables

$$\mathbf{Z}(x) = (x_1, x_2, x_1^2, x_1 x_2)^\top.$$

Your task is to consider **at least 3** the following estimation strategies:

1. **One-phase estimator with uniform i.i.d. sampling.**
2. **One-phase estimator with systematic sampling.**
3. **Stratified one-phase estimator with proportional uniform i.i.d. sampling.**
4. **Post-stratification using the same strata as the stratification estimator.** (Hint: this is Y_{reg} of a categorical variable). Use the plug-in variance estimator

For the stratification we partition the rectangular domain $\mathcal{D} = [0, 3] \times [0, 4]$ into four strata by bisecting each coordinate axis. The x_1-axis is split at its midpoint $x_1 = 1.5$ and the x_2-axis at its midpoint $x_2 = 2$. The four strata are then defined as

$$\mathcal{S}_1 = [0, 1.5) \times [0, 2), \tag{1}$$
$$\mathcal{S}_2 = [1.5, 3] \times [0, 2), \tag{2}$$
$$\mathcal{S}_3 = [0, 1.5) \times [2, 4], \tag{3}$$
$$\mathcal{S}_4 = [1.5, 3] \times [2, 4], \tag{4}$$

Thus, each strata proportion is $p_h = \frac{1}{4}$ for $h = 1, 2, 3, 4$.

```
get_stratum <- function(x1, x2) {
  ifelse(x2 < 2,
         ifelse(x1 < 1.5, "S1", "S2"), # lower half of x2
         ifelse(x1 < 1.5, "S3", "S4") # upper half of x2
         )
}
```

For each strategy, use a sample size of $n = 50$ and calculate the point estimate, the variance estimate, the confidence interval, and whether or not the true value is in the estimated confidence interval for that sample realization.

This is a Monte Carlo simulation, so we will repeat the procedure 500 times to calculate:

- the population variance, i.e., the variance of the replicate point estimates;
- the expectation of our variance estimator, i.e., the mean of the variance estimates;
- the empirical coverage rate, i.e., the mean of the binary variable indicating whether or not our confidence interval contained the true value.

Make a table with each estimation strategy as a row and columns for the true value, the expectation of the estimator, the population variance, the expectation of the variance estimator, and the empirical coverage rate.

Comment on the efficiency of the different approaches. Think about what you would expect to happen if we did this again for larger sample sizes of n.

Evaluation criteria

The examination is marked out of **20 points**. Assessment concerns the statistical reasoning, the consistency between sampling designs and estimators, the implementation in R, and the interpretation of results. Correct numerical results alone are insufficient without an explanation of how they were obtained.

1. Sampling exercise: post-storm remeasurement — 8 points

1. **Definition and justification of three probability sampling designs — 3 points.** The sampling units, selection procedures, and relevant selection probabilities are clearly specified. The designs are distinct, feasible, and justified in relation to the remeasurement objective.

2. **Estimators and uncertainty — 3 points.** Appropriate estimators are proposed for total affected stems and mean affected basal area, with explicit definitions and units. The uncertainty calculation is consistent with each sampling design. Required information and assumptions are identified.

3. **Visual presentation and practical recommendation — 2 points.** Sample realisations are presented clearly. Advantages and disadvantages are discussed in terms of precision, field effort, and feasibility. The recommendation is justified and understandable to the forest owner.

2. Estimation exercise: continuous spatial population — 12 points

1. **Implementation of at least three strategies — 3 points.** Sample selection is correctly implemented over the domain, with appropriate randomisation, stratification or post-stratification, and sampling or area weights where relevant.

2. **Point estimation and uncertainty — 4 points.** Estimates of the integral, variance estimates, and 95% confidence intervals are correctly calculated and explained. The variance method matches the sampling strategy; approximations and limitations are acknowledged.

3. **Monte Carlo evaluation — 3 points.** The 500 repetitions are implemented reproducibly. The comparison table correctly reports the mean estimate, empirical variance of estimates, mean estimated variance, and empirical coverage. The distinction between these quantities is understood.

4. **Interpretation and comparison — 2 points.** Conclusions about efficiency and coverage follow from the results. Differences between strategies are explained, together with the expected consequences of increasing the sample size.

Marking principles. Partial credit is awarded for sound reasoning even when a numerical or coding error occurs. The same propagated error is not penalised repeatedly. Code that runs does not compensate for an estimator or variance formula that is incompatible with the sampling design. Marks above 16 require both technical accuracy and strong interpretation; additional methods or lengthy reports do not, by themselves, justify a higher mark.

Individual verification of understanding. A short oral discussion will assess the student's understanding of essential choices and calculations in the submitted work. Questions will concern sampling, estimation, uncertainty, simulation, and interpretation, and may include a small modification of the code. This discussion will constitute a separate examination. Marks above 16 require strong individual understanding, in addition to accurate submitted results.